From discontinuity covers to certified adaptive quadrature

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Abstract

The covering criterion for Riemann integrability becomes a numerical certificate when each partition cell has a computable range enclosure. We analyze largest-radius bisection under this information model. Local sharpness outside a set with arbitrarily short interval covers ensures termination. An exact count of the associated threshold tree relates the number of queries to the scale and amplitude of unresolved ranges. If at most $O(2^{dm})$ cells at depth m are exceptional, with $0 < d < 1$, and the remaining ranges satisfy a Hölder bound of order α, the query complexity is $O(\varepsilon^{-\max\{1/\alpha, d/(1-d)\}})$. The regular contribution disappears for piecewise constant regions. For thresholds accumulating at $k^{-\beta}$, adaptive certificates require $\Theta(\varepsilon^{-1/\beta})$ cells, whereas uniform dyadic certificates require $\Theta(\varepsilon^{-(\beta+1)/\beta})$. A Cantor-set example also has matching adaptive bounds. Exact rational experiments illustrate these orders and the failure of a point-sampling error test on a narrow pulse.

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1 Introduction

For a bounded Riemann-integrable function $f : [0, 1] \to \mathbb{R}$, the computational objective is to return rational bounds

$$I(f) := \int_0^1 f(x)\,dx \in [A, B], \qquad \frac{B - A}{2} \le \varepsilon. \tag{1.1}$$

The midpoint then has absolute error at most ε. The information supplied to the algorithm consists of computable interval enclosures of f on dyadic cells.

The construction follows the covering proof of Riemann integrability. Intervals of small total length control the contribution of the discontinuities; continuity controls the remaining cells. For numerical use, this argument requires a procedure that encloses the range on each cell. Point values alone cannot supply that information on an unrestricted class of smooth functions, as Proposition 2.2 shows.

The discontinuity-covering criterion is the classical Lebesgue criterion in interval-cover form [1]. Global adaptation by selecting the largest local error is standard [2], as are verified interval sums [3]. Higher-order verified integration includes the Petras framework and its implementation in ball arithmetic [4]; Moczurad and Zgliczyński analyze its complexity for piecewise analytic functions [5]. The present analysis concerns the simpler interval-sum certificate. Its quantitative results are an exact threshold-tree count, geometric query bounds, and matching bounds for accumulating and fractal exceptional sets.

All complexity statements refer to calls to the specified range oracle. The lower bounds concern partitions using that oracle; a formula for the integral or stronger analytic information can lead to a different computational problem.

2 The covering criterion and point information

A partition of $[a,b]$, $a<b$, is a finite family of positive-length closed intervals with disjoint interiors and union $[a,b]$. Its mesh is the largest cell length. For bounded $f:[a,b]\to\mathbb{R}$, write

$$\operatorname{osc}(f,J)=\sup_{u,v\in J}|f(u)-f(v)|,\qquad \omega_f(x)=\inf_{r>0}\operatorname{osc}(f,(x-r,x+r)\cap[a,b]).$$

Endpoint continuity is relative to $[a,b]$. The discontinuity set is $D_f=\{x:\omega_f(x)>0\}$, and the Darboux gap is

$$U(f,P)-L(f,P)=\sum_{J\in P}|J|\operatorname{osc}(f,J). \tag{2.1}$$

Criterion 2.1 (Discontinuity covers)**.** *A bounded function $f:[a,b]\to\mathbb{R}$ is Riemann integrable if and only if, for every $\eta>0$, there are open intervals $(I_n)_{n\ge1}$ such that*

$$D_f\subseteq\bigcup_{n\ge1}I_n,\qquad \sum_{n\ge1}|I_n|<\eta. \tag{2.2}$$

Proof. For sufficiency, let $M=\sup|f|$ and suppose $M>0$. Given $\varepsilon>0$, choose a cover with total length below $\eta=\varepsilon/(4M)$. At each continuity point x, choose a relative open neighborhood V_x on which the oscillation is below $t=\varepsilon/[2(b-a)]$. The intervals $I_n\cap[a,b]$ and the neighborhoods V_x cover $[a,b]$. A finite subcover has a Lebesgue number $\delta>0$: every interval of length below δ lies in one of its members. Indeed, otherwise intervals of lengths tending to zero, each outside every single member, would have points converging along a subsequence to a covered point, a contradiction.

Take a partition of mesh below δ. Assign a cell to a selected exceptional interval whenever possible, and otherwise to a continuity neighborhood. The first group has total length below η and oscillation at most $2M$; the second has oscillation below t. Hence

$$U(f,P)-L(f,P)<2M\eta+t(b-a)=\varepsilon.$$

The case $M=0$ is immediate.

For necessity, fix $t,\eta>0$ and set $E_t=\{x:\omega_f(x)\ge t\}$. Choose a partition with Darboux gap below $t\eta/2$. Cells of oscillation at least t have total length below $\eta/2$ and contain every point of E_t except possibly partition endpoints. Enlarge those cells to open intervals of total length below $3\eta/4$, and cover all partition endpoints by intervals of total length below $\eta/4$. Thus E_t has an open interval cover of total length below η. Since $D_f=\bigcup_{k\ge1}E_{1/k}$, covers of these sets with lengths below $\eta2^{-k}$ establish (2.2). □

The proof uses a finite cover of the whole domain, combining exceptional intervals with continuity neighborhoods. It does not assume that D_f alone has a finite short cover; the discontinuities may be dense.

Proposition 2.2 (Finite point data)**.** *Consider a deterministic algorithm supplied only with values of an unknown smooth function $f:[0,1]\to[0,1]$. Suppose it makes finitely many evaluations on the zero-function transcript and returns an interval containing the integral of every admissible function consistent with that transcript. Its interval has width at least one.*

Proof. Let the queried points be $x_1,\dots,x_n$. For $A>0$, define

$$g_A(x)=\prod_{j=1}^{n}(1-e^{-A(x-x_j)^2}).$$

This smooth function lies in $[0,1]$ and vanishes at every queried point, so it produces the same adaptive transcript as zero. Moreover,

$$0 \le 1 - g_A(x) \le \sum_{j=1}^{n} e^{-A(x-x_j)^2}, \qquad \int_0^1 g_A(x)\,dx \ge 1 - \frac{3n}{\sqrt{A}}.$$

The latter estimate follows by substitution and $\int_{\mathbb{R}} e^{-u^2}\,du \le 2 + 2\int_1^\infty e^{-u}\,du < 3$. The returned interval must contain zero and integrals arbitrarily close to one. If $n = 0$, the two constant functions suffice. □

This obstruction assumes no derivative bound, modulus of continuity, or range enclosure. Such additional information can make point-based certification possible.

3 Range enclosures and adaptive bisection

Let $\mathcal{D}_m$ be the 2^m closed dyadic cells of length 2^{-m} in $[0,1]$. A dyadic partition is the leaf set of a finite binary subdivision tree.

Definition 3.1 (Range oracle)**.** A range oracle is a fixed map on dyadic cells that returns, in finite time, a nonempty bounded rational interval $\widehat{R}(J) = [\widehat{l}_J, \widehat{u}_J]$ satisfying

$$f(x) \in \widehat{R}(J) \qquad (x \in J). \tag{3.1}$$

Define inherited enclosures on the whole dyadic tree by

$$R([0,1]) = \widehat{R}([0,1]), \qquad R(J') = \widehat{R}(J') \cap R(J) \tag{3.2}$$

when J' is a child of J. Write $R(J) = [l_J, u_J]$, $w(J) = u_J - l_J$, and $W = w([0,1])$.

The intersection is nonempty because both ranges contain every value of f on J'. Inheritance enforces nesting even when the raw oracle is not monotone. The ranges are pointwise ranges, including endpoints. A query is one call for a raw enclosure.

For a partition P, define

$$A(P) = \sum_{J\in P} |J| l_J, \qquad B(P) = \sum_{J\in P} |J| u_J, \tag{3.3}$$

$$Q(P) = \frac{A(P)+B(P)}{2}, \qquad E(P) = \frac{B(P)-A(P)}{2} = \sum_{J\in P} e(J), \tag{3.4}$$

$$e(J) = \frac{|J| w(J)}{2}.$$

Algorithm 3.2 (Largest-radius bisection)**.** Given a positive rational tolerance ε, query the root and initialize the sums in (3.3). While $E(P) > \varepsilon$, bisect a leaf of largest $e(J)$, query its two children, apply (3.2), and update the sums. Return $[A(P), B(P)]$ and its midpoint when the radius test holds.

The implementation breaks ties by smaller depth, then smaller left endpoint. An interrupted run retains a valid enclosure, but certifies the requested tolerance only if its radius test has passed.

Proposition 3.3 (Validity and bookkeeping)**.** *If f is Riemann integrable, every computed partition satisfies*

$$I(f) \in [A(P), B(P)], \qquad |I(f) - Q(P)| \le E(P).$$

Refinement gives nested integral enclosures. Each child has radius at most half its parent's radius. After s bisections there are $s+1$ leaves and $2s+1$ queries.

Proof. Integrate $l_J \le f \le u_J$ on each cell and add. Each child has half the parent's length and a contained range. The children's lower contributions therefore sum to at least the parent's lower contribution; their upper contributions sum to at most its upper contribution. This proves nesting and the local radius bound. Every split adds one leaf and two queries, following the initial root query. □

One possible oracle uses a global range on exceptional cells and a known Hölder estimate on regular cells. If $|f(x) - f(y)| \le L|x - y|^\alpha$ on J, then

$$f(J) \subseteq [f(c) - L(|J|/2)^\alpha,\ f(c) + L(|J|/2)^\alpha], \qquad c = \mathrm{mid}(J). \tag{3.5}$$

For use in Definition 3.1, these endpoints must be enclosed outward by rational numbers, including evaluation errors in $f(c)$, L, and the power. The examples below instead use explicit exact rational routines.

4 Termination and the threshold-tree profile

Definition 4.1 (Local sharpness)**.** The oracle is locally sharp at a continuity point x if, for every $t > 0$, some relative open neighborhood V_x satisfies

$$\operatorname{diam} \widehat{R}(J) < t \qquad \text{for every dyadic cell } J \subseteq V_x.$$

Exact ranges have this property at continuity points, although computing them may be difficult. A valid oracle whose ranges remain wide need not have it.

Lemma 4.2. *Suppose $D_f \subseteq E \subseteq [0,1]$, where E admits open interval covers of arbitrarily small total length. If the oracle is locally sharp at every point outside E, then $E(\mathcal{D}_m) \to 0$.*

Proof. If $W = 0$, the conclusion is immediate. Otherwise cover E by intervals of total length below ε/W and choose sharpness neighborhoods with width bound ε outside E. A finite joint subcover has a positive mesh threshold, as in Criterion 2.1. On a sufficiently fine dyadic grid, assign cells to exceptional intervals whenever possible and otherwise to sharpness neighborhoods. Their inherited widths are at most W and below ε, respectively. Thus

$$E(\mathcal{D}_m) < \tfrac{1}{2}W(\varepsilon/W) + \tfrac{1}{2}\varepsilon = \varepsilon.$$

□

Theorem 4.3 (Finite termination)**.** *Under the hypotheses of Lemma 4.2, Algorithm 3.2 terminates at every positive rational tolerance, provided the run is not interrupted.*

Proof. Criterion 2.1 gives integrability. If the root already meets the tolerance, there is nothing to prove. Otherwise choose $m \ge 1$ with $E(\mathcal{D}_m) \le \varepsilon$. The positive-radius internal nodes of this comparison tree form a finite nonempty set. Let $v > 0$ be their smallest radius.

As long as one such node remains unsplit, it or an ancestor is an active leaf with radius at least v. The greedy rule therefore selects a node of radius at least v. Since $e(J) \le (W/2)2^{-\operatorname{depth}(J)}$, only finitely many nodes in the whole dyadic tree have such a radius. Each can be split only once. Consequently all positive-radius internal nodes of the comparison tree are eventually split unless the algorithm has stopped earlier.

At that point, every active leaf above depth m has radius zero, and every other leaf refines a depth-m cell. Zero-radius nodes have only zero-radius descendants. Nesting therefore places the current integral enclosure inside the depth-m enclosure, whose radius is at most ε. □

For the fixed inherited ranges, define the scale-and-amplitude profile

$$M_m(t) = \#\{J \in \mathcal{D}_m : w(J) > t\}, \qquad t > 0. \tag{4.1}$$

Unlike a count of cells meeting the discontinuity set, this profile can be useful even when that set is dense.

Proposition 4.4 (Exact threshold-tree count)**.** *For $\tau > 0$, split every node with $e(J) > \tau$ and let P_τ be the resulting partition. It is finite, and*

$$N(\tau) := \#P_\tau = 1 + \sum_{m \geq 0} M_m(2^{m+1}\tau), \qquad E(P_\tau) \leq \tau N(\tau). \tag{4.2}$$

If $\tau N(\tau) \leq \varepsilon$, Algorithm 3.2 stops with at most $N(\tau)$ leaves and $2N(\tau) - 1$ queries.

Proof. The set $\{J : e(J) > \tau\}$ is ancestral, since a parent's radius is at least twice its child's. It is finite by the depth bound used above. Its nodes are exactly the internal nodes of the threshold tree. A finite full binary tree has one more leaf than internal nodes, and at depth m the condition is equivalent to $w(J) > 2^{m+1}\tau$. Every remaining leaf has radius at most τ, proving (4.2).

Until all threshold-tree internal nodes are split, an active ancestor of an unsplit node has radius above τ. Greedy refinement therefore splits only threshold-tree nodes. Once they have all been split, the tolerance holds; the run may have stopped earlier. □

The threshold τ is used only in the analysis. The algorithm monitors $E(P)$ directly.

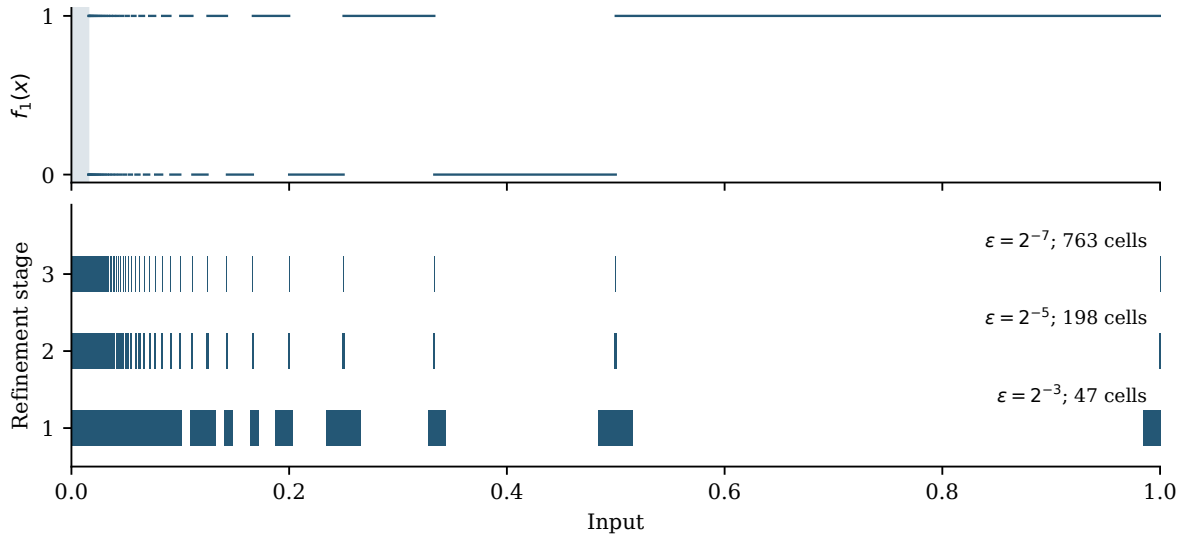


Figure 1: Alternating thresholds at k^{-1}. The upper panel shows the first 64 constant branches and the unresolved tail near zero. The lower panel shows exceptional leaves at three tolerances. Their total length is twice the certificate radius.

5 Geometric query bounds

Classify each dyadic cell as regular or exceptional. The classification may be conservative and need not be nested across levels. Suppose that, for fixed parameters

$$0 < \alpha \leq 1, \quad a \geq 0, \quad b > 0, \quad C \geq 1, \quad 0 \leq d < 1,$$

the inherited radii satisfy

$$e(J) \le a|J|^{1+\alpha} \qquad \text{on regular cells,} \tag{5.1}$$
$$e(J) \le b|J| \qquad \text{on exceptional cells,} \tag{5.2}$$
$$n_m := \#\{J \in \mathcal{D}_m : J \text{ exceptional}\} \le C2^{dm}. \tag{5.3}$$

The count n_m must include all cells left unresolved by the actual test. Counting only cells meeting the true discontinuity set may underestimate the work of a conservative oracle.

These assumptions already imply integrability, because $E(\mathcal{D}_m) \le a2^{-\alpha m} + Cb2^{-(1-d)m} \to 0$ and the true Darboux gap is at most twice this radius.

Theorem 5.1 (Explicit threshold bound)**.** *Assume* (5.1)–(5.3) *with* $0 < d < 1$*, and set*

$$q = \frac{1}{1+\alpha}, \qquad A_\alpha = 2a^q, \qquad B_d = \frac{C2^d}{2^d - 1} b^d.$$

For every $\tau > 0$*,*

$$N(\tau) \le 1 + A_\alpha \tau^{-q} + B_d \tau^{-d}, \tag{5.4}$$
$$E(P_\tau) \le \tau + A_\alpha \tau^{\alpha/(1+\alpha)} + B_d \tau^{1-d}. \tag{5.5}$$

Thus a positive rational choice satisfying

$$\tau \le \min\left\{ \frac{\varepsilon}{3}, \left(\frac{\varepsilon}{3A_\alpha}\right)^{(1+\alpha)/\alpha}, \left(\frac{\varepsilon}{3B_d}\right)^{1/(1-d)} \right\} \tag{5.6}$$

gives at most $2N(\tau) - 1$ *queries. The term containing* A_α *is omitted when* $a = 0$*.*

Proof. A regular node at depth m can have $e(J) > \tau$ only if $a2^{-(1+\alpha)m} > \tau$. Summing at most 2^m nodes at each eligible depth gives at most $2(a/\tau)^q$ such nodes; there are none when $a = 0$.

An exceptional node can exceed the threshold only if $b2^{-m} > \tau$. If $b > \tau$, summing $C2^{dm}$ over $m < \log_2(b/\tau)$ gives fewer than

$$\frac{C2^d}{2^d - 1}(b/\tau)^d$$

nodes. If $b \le \tau$, there are none. Proposition 4.4 gives (5.4); multiplication by τ gives (5.5). Each term in the latter is at most $\varepsilon/3$ under (5.6). □

Corollary 5.2 (Tolerance complexity)**.** *Let* $Q(\varepsilon)$ *count the queries of Algorithm 3.2. Under* (5.1)–(5.3)*, for* $0 < d < 1$*,*

$$Q(\varepsilon) = \begin{cases} O\left(\varepsilon^{-\max\{1/\alpha, d/(1-d)\}}\right), & a > 0, \\ O\left(\varepsilon^{-d/(1-d)}\right), & a = 0, \end{cases} \qquad \varepsilon \downarrow 0. \tag{5.7}$$

If $n_m \le C$ *at every depth, the corresponding bounds are* $O(\varepsilon^{-1/\alpha})$ *when* $a > 0$ *and* $O(\log(1/\varepsilon))$ *when* $a = 0$*.*

Proof. For $a > 0$, put $\kappa = \max\{q, d\} < 1$. Equations (5.4)–(5.5) give $N(\tau) = O(\tau^{-\kappa})$ and $E(P_\tau) = O(\tau^{1-\kappa})$. Taking a sufficiently small constant multiple of $\varepsilon^{1/(1-\kappa)}$ proves the claim, since $\kappa/(1-\kappa) = \max\{1/\alpha, d/(1-d)\}$. When $a = 0$, take $\kappa = d$.

For $d = 0$, the exceptional contribution is at most $C \max\{0, \lceil \log_2(b/\tau) \rceil\}$, so

$$N(\tau) \le 1 + 2(a/\tau)^q + C \max\{0, \lceil \log_2(b/\tau) \rceil\}.$$

If $a > 0$, the power dominates the logarithm. If $a = 0$, the choice $\tau = \varepsilon^2$ gives $\tau N(\tau) \le \varepsilon$ for sufficiently small ε and logarithmic growth in $N(\tau)$. □

The case $d = 0$ requires a uniform bound on the cell counts; box dimension zero alone is insufficient. For a uniform partition with $N = 2^m$ cells,

$$E(\mathcal{D}_m) \le aN^{-\alpha} + CbN^{-(1-d)}. \tag{5.8}$$

When $a = 0$, this gives a sufficient uniform count $O(\varepsilon^{-1/(1-d)})$. The following examples show where these different adaptive and uniform orders are attained.

A heap adds $O(N \log N)$ comparisons for N leaves. Query evaluation, rational arithmetic, and integer bit lengths remain separate costs.

6 Examples and matching certificate bounds

For the switching examples, the geometric oracle returns $[0, 1]$ on every positive-length closed interval meeting the specified exceptional set and the exact constant elsewhere. Its radius is half the total length of exceptional leaves. The oracle is defined on arbitrary interval partitions as well as dyadic ones.

6.1 Accumulating thresholds

For $\beta > 0$, let

$$E_\beta = \{0\} \cup \{k^{-\beta} : k \ge 1\}, \qquad f_\beta(x) = \begin{cases} 1, & x > 0 \text{ and } \lfloor x^{-1/\beta} \rfloor \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases} \tag{6.1}$$

The discontinuity set is $E_\beta \setminus \{1\}$. We retain 1 in the exceptional set to simplify the interval test. A countable set has arbitrarily short interval covers, so Criterion 2.1 applies. Summing the constant branches gives

$$I(f_\beta) = \sum_{k \text{ odd}} \left(k^{-\beta} - (k+1)^{-\beta}\right) = \sum_{k \ge 1} (-1)^{k+1} k^{-\beta} =: \eta(\beta). \tag{6.2}$$

The remaining tail after the first K branches lies in $[0, (K+1)^{-\beta}]$, so its contribution tends to zero. In particular, $I(f_1) = \log 2$.

For arbitrary real β, the analysis assumes access to the stated range information. The executable examples use $\beta = 1, 2$, for which finite exact tests are given in Appendix B.

Lemma 6.1. *For $d = 1/(\beta+1)$, the number of depth-m cells meeting E_β satisfies*

$$n_m \asymp 2^{dm} \quad (m \to \infty), \qquad n_m \le 6\, 2^{dm} \quad (m \ge 0). \tag{6.3}$$

Proof. Put $h = 2^{-m}$ and $K = \lceil h^{-1/(\beta+1)} \rceil$. The first K thresholds meet at most $2K$ closed cells. The remaining thresholds and zero lie in $[0, (K+1)^{-\beta}]$, meeting at most $1 + h^{-1}(K+1)^{-\beta}$ cells. Thus $n_m \le 3h^{-1/(\beta+1)} + 3 \le 6h^{-1/(\beta+1)}$.

For the lower bound, take $K = \lfloor (\beta/(4h))^d \rfloor - 1$, positive for small h. For $1 \le k < K$,

$$k^{-\beta} - (k+1)^{-\beta} \ge \beta(k+1)^{-\beta-1} \ge \beta(K+1)^{-\beta-1} \ge 4h.$$

These K thresholds cannot share a cell of length h, giving $n_m \ge K \asymp h^{-d}$. □

Theorem 6.2 (Accumulating-threshold complexity)**.** *For the geometric oracle of f_β,*

$$Q(\varepsilon) = O(\varepsilon^{-1/\beta}), \qquad E(\mathcal{D}_m) \asymp N^{-\beta/(\beta+1)}, \quad N = 2^m.$$

The minimum number of leaves among all interval partitions with radius at most ε is $\Theta(\varepsilon^{-1/\beta})$. The same order holds for dyadic partitions and is attained by greedy refinement. Full uniform dyadic partitions require $\Theta(\varepsilon^{-(\beta+1)/\beta})$ cells.

Proof. Corollary 5.2, with $a = 0$, $b = 1/2$, and $d = 1/(\beta + 1)$, gives the adaptive upper bound. The uniform radius is exactly $hn_m/2$, so Lemma 6.1 gives its two-sided order.

For the lower bound, let a partition have $N \geq 1$ leaves. Assign each of the first $2N$ thresholds to one exceptional leaf containing it. Their minimum separation is at least $\Delta = \beta(2N+1)^{-\beta-1}$. If r_J points are assigned to a leaf J, then $|J| \geq (r_J - 1)\Delta$. There are at most N nonempty groups and $\sum_J r_J = 2N$. Their total length is therefore at least $N\Delta$, whence

$$E(P) \geq \frac{\beta}{2} N(2N+1)^{-\beta-1} \geq \frac{\beta}{2\,3^{\beta+1}} N^{-\beta}. \tag{6.4}$$

This applies to arbitrary partitions; the greedy dyadic construction supplies the matching upper bound. Inverting the uniform radius estimate and rounding the count to a power of two proves the uniform order. □

For a mixed example, take $g(x) = f_1(x) + x^2/8$, with integral $\log 2 + 1/24$. On a regular cell $J = [s, t]$, use $[c + s^2/8, c + t^2/8]$, where c is the constant branch. On an exceptional cell use $[s^2/8, 1 + t^2/8]$. These nested ranges satisfy (5.1)–(5.2) with $\alpha = 1$, $a = 1/8$, and $b = 5/8$. The adaptive query bound is $O(\varepsilon^{-1})$. The switching part gives the lower uniform radius order $N^{-1/2}$, and (5.8) gives the upper order, so uniform certificates require $\Theta(\varepsilon^{-2})$ cells.

6.2 A Cantor exceptional set

Let C be the middle-thirds Cantor set. Define f_C to be one on the open gaps removed at odd generations and zero on the even-generation gaps and on C. Every retained interval contains later gaps of both parities. Hence both values occur arbitrarily close to every point of C, and $D_{f_C} = C$.

The 2^k stage-k retained intervals have total length $(2/3)^k$. Enlarging them slightly gives the covers required by Criterion 2.1. Gap summation yields

$$I(f_C) = \sum_{j \geq 0} \frac{2^{2j}}{3^{2j+1}} = \frac{3}{5}. \tag{6.5}$$

The omitted contribution is bounded by the length of the retained set, which tends to zero.

An exact geometric oracle is finite. If a cell lies strictly inside the current middle third, its generation determines the constant value. If it lies in a retained left or right third, rescale that third to $[0, 1]$ and repeat. Otherwise the cell meets C and receives $[0, 1]$. Each rescaling triples its positive length, so the procedure takes $O(1 + \log(1/|J|))$ iterations.

Proposition 6.3 (Measure-based lower bound)**.** *Let $E \subseteq [0, 1]$ be compact and let $f : [0, 1] \to [0, 1]$ be Riemann integrable and constant, with rational value, on each component of $[0, 1] \setminus E$. Suppose a Borel probability measure ν, supported on E, satisfies*

$$\nu(J) \leq K|J|^d \qquad (0 < d < 1) \tag{6.6}$$

for every positive-length closed interval J. With $[0, 1]$ on leaves meeting E and the exact constant elsewhere, every N-leaf partition satisfies

$$E(P) \geq \tfrac{1}{2} K^{-1/d} N^{-(1-d)/d}. \tag{6.7}$$

Proof. Let the exceptional leaves have total length δ and number $r \leq N$. They cover E, so Hölder's inequality gives

$$1 \leq \sum_{J \text{ exceptional}} \nu(J) \leq K \sum_{J \text{ exceptional}} |J|^d \leq K r^{1-d} \delta^d \leq K N^{1-d} \delta^d.$$

The radius is $\delta/2$. □

Corollary 6.4 (Cantor certificates)**.** *For $d = \log 2/\log 3$, the minimum leaf count for f_C in its geometric enclosure model is $\Theta(\varepsilon^{-d/(1-d)})$, attained by greedy dyadic refinement up to constants. Uniform dyadic certificates need at most $O(\varepsilon^{-1/(1-d)})$ cells.*

Proof. For mesh h, choose $k = \lceil \log_3(1/h) \rceil$. Each of the 2^k retained intervals has length at most h and meets at most three closed grid cells. Thus $n_m \le 3 \cdot 2^k \le 6h^{-d}$. Corollary 5.2 and (5.8) give the upper bounds.

Let ν be the law of $\sum_{n\ge1} 2B_n 3^{-n}$ for independent fair binary digits B_n. Each stage-k interval has mass 2^{-k}. Given an interval of length ℓ, choose $k \ge 0$ with $3^{-(k+1)} < \ell \le 3^{-k}$. It meets at most two stage-k intervals, because spanning three would require length greater than 3^{-k}. Hence

$$\nu(J) \le 2 \cdot 2^{-k} = 2(3^{-k})^d \le 4\ell^d.$$

Proposition 6.3, with $K = 4$, gives the matching lower bound. □

7 Exact rational experiments

The accompanying implementation uses Python's `Fraction` for interval endpoints, ranges, integral sums, priorities, and stopping decisions. Floating conversion is confined to plotting. The six examples are a single jump $\mathbf{1}_{\{x>1/3\}}$, a narrow pulse, f_1, f_2, $f_1 + x^2/8$, and f_C. Each is run at five tolerances. Every returned enclosure contains an independently constructed reference interval and meets the requested radius.

Table 1: Selected certificates. Uniform counts are full-grid cardinalities. The adaptive query count is twice the leaf count minus one.

Integrand	ε	Adaptive leaves	Queries	Uniform cells
One jump	2^{-20}	20	39	524,288
Narrow pulse	2^{-20}	28	55	1,048,576
f_1	2^{-10}	5,826	11,651	2,097,152
f_2	2^{-13}	725	1,449	1,048,576
$f_1 + x^2/8$	2^{-10}	5,928	11,855	2,097,152
Cantor gaps	2^{-6}	4,293	8,585	131,072

Table 1 and Figure 2 compare partitions using the same range specification. Full uniform sums are evaluated by aggregating regular subtrees, without allocating every uniform leaf. The full-grid cardinality and the number of calls made by this aggregation routine are stored separately. These are comparisons of certificate sizes, rather than runtime measurements.

7.1 A narrow pulse and point sampling

Let $p(x) = \mathbf{1}_{(3699/10000,3701/10000)}(x)$. Its integral is $1/5000$, but its values at $0, 1/4, 1/2, 3/4, 1$ are all zero. The single-panel Simpson approximation S and its two-panel refinement S_2 are therefore both zero, as is the difference estimate $|S_2 - S|/15$. A stopping test based only on these observations can accept zero.

The range oracle detects the two pulse endpoints. At tolerance 2^{-20} it uses 55 queries and returns

$$\int_0^1 p(x)\,dx \in \left[\frac{13}{65536}, \frac{105}{524288}\right], \qquad E = \frac{1}{1048576}. \tag{7.1}$$

The example isolates the extra information provided by range enclosures. Small quadrature differences can fail as error indicators for reasons discussed by Gonnet [2].

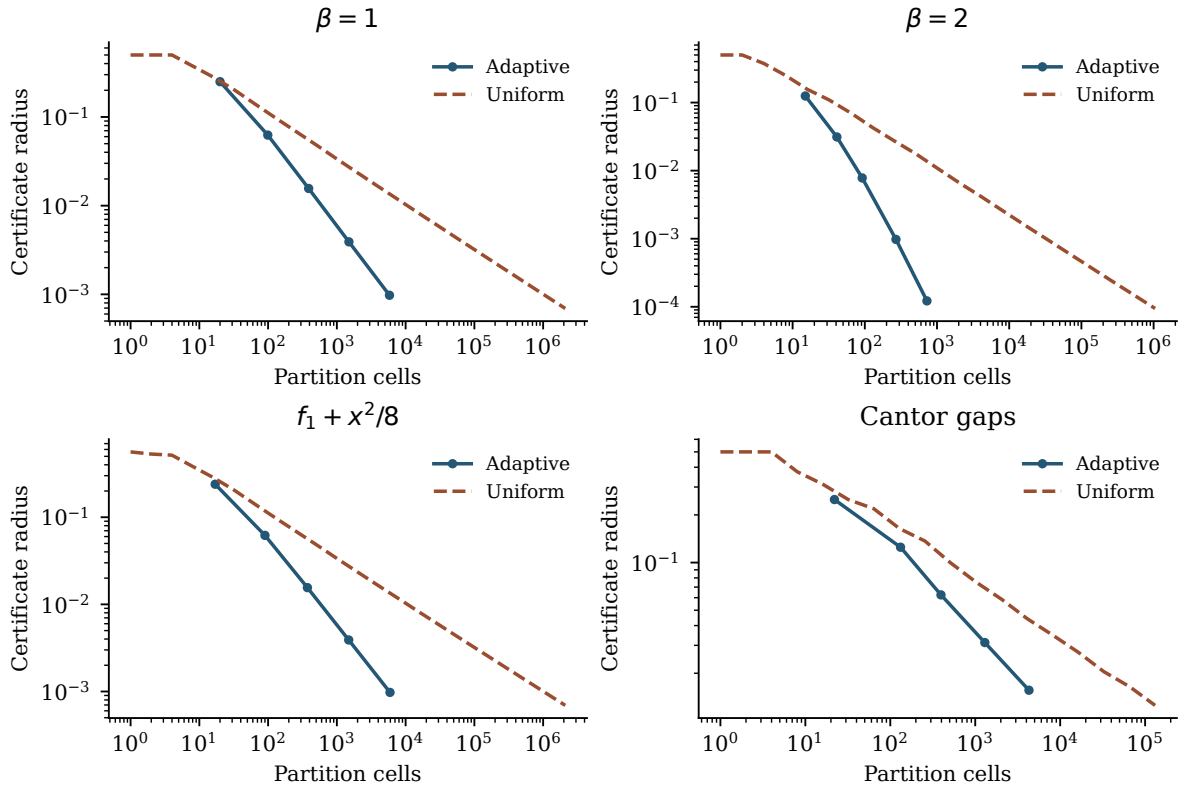


Figure 2: Certificate radius against partition size. Adaptive points use the five recorded tolerances for each example; uniform curves use complete dyadic grids evaluated by exact aggregation.

7.2 Reproduction

Running `python reproduce.py` regenerates the 30 certificates, the uniform-grid records, and both figures. The reference intervals for f_1 and f_2 use the independent series construction in Appendix A. The script also checks the threshold tests against explicit enumeration, the Cantor test against finite retained-interval covers, aggregated sums against cell-by-cell sums, and sampled values against the stated ranges. The exact inputs, outputs, and check totals are included in the data directory.

8 Conclusion

Computable range enclosures turn the discontinuity-cover argument into a certified adaptive procedure. The threshold-tree profile counts the cells whose local radii exceed a given scale, while the geometric estimates relate that count to exceptional-set coverage and regularity away from it. Accumulating thresholds and Cantor gaps give matching adaptive orders within the specified certificate model.

The effective input is the range oracle, not Riemann integrability alone. Its quality determines whether refinement terminates and how much work is required. For dense discontinuity sets, a geometric count may be uninformative even when the amplitude profile remains useful. Extending the bounds in that direction requires quantitative control of range widths as well as discontinuity locations.

A Reference enclosures for the switching integrals

For $\beta = 1, 2$, put $a_k = (k+1)^{-\beta}$ and $\Delta a_k = a_k - a_{k+1}$. Elementary integration gives

$$a_k = \int_0^1 t^k w_\beta(t)\, dt, \qquad w_1(t) = 1, \quad w_2(t) = -\log t.$$

Both weights are nonnegative and integrate to one. Consequently,

$$0 \le \Delta^n a_0 = \int_0^1 (1-t)^n w_\beta(t)\, dt \le 1.$$

Summing the positive geometric series gives

$$\sum_{n\ge 0} \frac{\Delta^n a_0}{2^{n+1}} = \int_0^1 \frac{w_\beta(t)}{1+t}\, dt = \eta(\beta). \tag{A.1}$$

The tail after M terms is at most $2^{-M}\int_0^1 w_\beta = 2^{-M}$. For the final identity, integrate the finite geometric identity $\sum_{k=0}^{K-1}(-t)^k = (1-(-t)^K)/(1+t)$. The absolute remainder is at most $\int_0^1 t^K w_\beta(t)\, dt = (K+1)^{-\beta} \to 0$.

The first M terms in (A.1) are rational, so they give a lower bound and adding 2^{-M} gives an upper bound. The implementation takes $M = 100$. The other references are $2/3$, $1/5000$, and $3/5$, with $1/24$ added for the quadratic background.

B Finite interval tests and uniform aggregation

For $a > 0$, a cell $[a, b]$ meets E_β precisely when

$$\left\lceil b^{-1/\beta} \right\rceil \le \left\lfloor a^{-1/\beta} \right\rfloor. \tag{B.1}$$

At $a = 0$ it is exceptional. For $\beta = 1$ the comparisons are rational. For $\beta = 2$, compute $\lfloor\sqrt{p/q}\rfloor$ as isqrt($\lfloor p/q \rfloor$) and use an exact equality check for the ceiling. The query therefore uses finitely many integer operations. The Cantor routine uses rational comparisons with $1/3$ and $2/3$, followed by the rescaling described in Section 6.2.

Independent checks enumerate thresholds through $k = N + 1$, leaving the tail in the first grid cell, and compare Cantor classifications with a retained-interval cover finer than the grid. Every grid cell disjoint from C lies inside a gap already removed at that stage.

For the mixed model, a regular block $[a, b]$ with n uniform subcells of width h and constant switching value c has lower sum

$$c(b-a) + \frac{h}{8}\left(na^2 + ah\, n(n-1) + h^2 \frac{n(n-1)(2n-1)}{6} \right).$$

Its upper sum exceeds this by $h(b^2 - a^2)/8$. These identities allow exact aggregation without changing the uniform certificate.

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